

Blockchain-enabled Intelligent Scheme for BankSEC: Secure Smart Banking Environment

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Abstract

The rapid evolution of digital banking presents significant challenges related to security, trust, and transparency in conventional smart banking systems, which often depend on centralized infrastructure and are consequently susceptible to fraud, identity theft, and data breaches. To address above challenges, we propose an Intelligent scheme for secure Smart Banking Environment based on blockchain, named BankSEC, which aims to eliminate single points of failure and enhance trust in digital transactions. The proposed system stores all financial activities on a decentralized, tamper-resistant ledger, ensuring integrity, transparency, and auditability. Biometric authentication such as fingerprint recognition is used instead of passwords to ensure strong and reliable identity validation. A Decentralized Identity (DID) mechanism further enables users to control their verified identity and minimize repeated KYC checks. Artificial intelligence is integrated to analyze transaction patterns in real time and detect suspicious behavior, thereby preventing fraud before it occurs. Smart contracts automate secure payments, P2P transfers, and loan processes, ensuring accuracy, efficiency, and transparency without third-party intervention. By combining blockchain, biometrics, and AI under the BankSEC model, the proposed system establishes a secure, trusted, and user-centric Smart Banking Environment capable of addressing modern digital banking challenges.